

Super critical fluid chromatography



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INTRODUCTION

Supercritical Fluid Chromatography (SFC) is an advanced chromatographic technique that combines the advantages of Gas Chromatography (GC) and High Performance Liquid Chromatography (HPLC). It is mainly used for the separation and purification of low to moderate molecular weight compounds, especially thermally labile substances.

Historical Background

1. Supercritical Fluid Chromatography was first proposed in 1958 by J. Lovelock.
2. It was first practically used by Klesper in 1962.
3. The technique combines features of both HPLC and GC.
4. It is widely used for analysis and purification of thermally unstable molecules.

DEFINITION

Supercritical Fluid Chromatography (SFC) is a high-efficiency separation technique that uses a supercritical fluid, usually carbon dioxide, as the mobile phase to combine the advantages of gas chromatography and liquid chromatography.

IMPORTANT TERMS

1. Critical Temperature

The highest temperature at which a substance can exist as a liquid.

2. Critical Pressure

The minimum pressure required to liquefy a gas at its critical temperature.

3. SFC Temperature

For every substance, there exists a temperature above which it can no longer exist as a liquid, regardless of the pressure applied.

4. SFC Pressure

For every substance, there exists a pressure above which it can no longer exist as a liquid, irrespective of temperature increase.

5. Triple Point

The point at which the three states of matter coexist at constant temperature and pressure.

PRINCIPLE OF SUPERCRITICAL FLUID CHROMATOGRAPHY

Supercritical Fluid Chromatography is based on the principles of both adsorption and partition chromatography. It combines the properties of Gas Chromatography and High Performance Liquid Chromatography.

In SFC:

- The sample is carried through the separating column by a supercritical fluid.
- Components of the mixture separate into distinct bands.
- Separation depends on the interaction between analytes and the stationary phase.
- As analytes leave the column, detectors identify and quantify them.

Key Features

- High diffusion rate similar to gases
- Solvent strength similar to liquids
- Faster separation with improved efficiency
- Suitable for thermally sensitive compounds

SUPERCRITICAL FLUIDS USED IN SFC

Carbon dioxide and water are the most commonly used supercritical fluids.

Carbon dioxide is the preferred mobile phase because of its:

- Low toxicity
- Low cost
- Easy availability
- Low critical temperature

Nitrogen and compressed air can also exist as supercritical fluids above their critical pressure and temperature.

CHOICE OF SUPERCRITICAL FLUID SOLVENT

An ideal supercritical fluid should possess the following properties:

- Good solubilizing and penetrating ability
- Chemically inert toward analytes
- Easy separation from products
- Low cost
- Non-toxic nature
- Easily available

PROPERTIES OF SUPERCRITICAL FLUIDS

Supercritical fluids possess unique physical and chemical properties that make them highly useful in chromatography.

Important Properties

1. There is no surface tension because no liquid-gas boundary exists.
2. Solubility increases with increasing fluid density.
3. Density increases with pressure; therefore, solubility also increases with pressure.
4. At constant density, solubility increases with temperature.
5. Supercritical fluids possess:
 - High diffusivity
 - Low viscosity
 - Better penetration ability

INSTRUMENTATION OF SUPERCRITICAL FLUID CHROMATOGRAPHY

The major components of SFC instrumentation include:

1. Mobile Phase
2. Stationary Phase
3. Pump
4. Injection System
5. Oven

6. Column
 7. Detector
 8. Back Pressure Device (Restrictor)
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1. MOBILE PHASE

Several fluids may be used as the mobile phase in SFC, but carbon dioxide is most commonly used.

Advantages of CO₂

- Cheap
- Non-flammable
- Easily removable
- Environmentally safer

Limitation

Carbon dioxide cannot efficiently elute highly polar or ionic compounds.

Solution

This limitation is overcome by adding modifier fluids such as:

- Alcohols
 - Cyclic ethers
 - Acetonitrile
 - Chloroform
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2. STATIONARY PHASE

The stationary phases used in SFC are similar to those used in GC and LC, with slight modifications.

Types of Stationary Phases

Silica and Alumina

- Useful for non-polar compounds
- May cause irreversible adsorption of polar compounds

Polar Stationary Phases

Polysiloxanes

- Stable
- Flexible Si–O bond
- Good diffusion properties

Polymethyl Siloxanes

- Improve efficiency in separating closely related polar analytes

Cyanopropyl Polysiloxanes

- Useful for compounds containing carboxylic acid groups (–COOH)

3. PUMPS

Pumps are used to maintain and control the flow of the mobile phase. Flow control is an important function in SFC systems.

Types of Pumps

Reciprocating Pumps

- Commonly used in packed column SFC
- Allow easy blending of mobile phases
- Useful for adding modifier fluids

Working Principle

A reciprocating pump works by moving a piston back and forth inside a cylinder.

4. OVEN

A thermostatically controlled column oven is required for precise temperature control of the mobile phase and column. Proper temperature control ensures:

- Stable supercritical conditions
 - Better reproducibility
 - Improved separation efficiency
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5. BACK PRESSURE DEVICE (RESTRICTOR)

The restrictor maintains the desired pressure inside the column using a pressure-adjustable diaphragm or controlled nozzle.

Functions

- Maintains supercritical conditions throughout separation
- Controls pressure inside the system

- Prevents phase changes
 - Often heated to avoid clogging
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6. INJECTORS

Injectors introduce the sample into the chromatographic system.

Capillary SFC

- Small sample volumes
- Pneumatically driven valves used

Packed Column SFC

- Conventional injection valves used
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7. COLUMNS

Two major analytical columns are used in SFC:

Packed Columns

- Contain small deactivated particles
- Stationary phase adheres to particles
- Usually made of stainless steel

Capillary Columns

- Open tubular columns
- Narrow internal diameter
- Made of fused silica

- Stationary phase bonded to inner wall
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8. DETECTORS

Most detectors used in GC and HPLC can also be used in SFC.

Common Detectors

Flame Ionization Detector (FID)

- Highly sensitive
- Commonly used in SFC

UV Detector

- Used for UV-absorbing compounds
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APPLICATIONS OF SUPERCRITICAL FLUID CHROMATOGRAPHY

Supercritical Fluid Chromatography has wide industrial and pharmaceutical applications.

Major Applications

- Separation of chiral molecules
- Analysis of natural products
- Drug analysis and purification
- Food analysis
- Polymer analysis
- Separation of thermally unstable compounds

- Pharmaceutical quality control
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ADVANTAGES OF SFC

- Faster analysis compared to HPLC
 - High efficiency and resolution
 - Reduced solvent consumption
 - Environmentally safer
 - Suitable for thermally labile compounds
 - Easy recovery of analytes
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LIMITATIONS OF SFC

- High instrumentation cost
 - Requires high pressure systems
 - Limited suitability for highly polar compounds
 - Complex pressure control system
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